PySR Warm-Start (Hybrid) vs. Cold PySR: Trajectory Analysis Report Across 5 Seeds

Generated by `generate_figures.py` (exp3 trajectory block)

August 16, 2026

Abstract

This report was generated by running `generate_figures.py` (`--experiment exp3`) against the five real `exp3_nguyen12_seed*.json` trajectory logs (seeds 42, 99, 123, 777, 2024; 12 Nguyen-12 equations each, 60 equation–seed pairs total). Every figure and number below is a direct, unedited output of the pipeline on that data.

Headline result. On **final loss at whatever iteration each arm's run happened to stop**, the two arms are statistically indistinguishable in practice: 39/60 equation–seed pairs end in an exact tie (both arms converge to the same near-machine-precision loss, $\sim 10^{-15}$–10^{-16}), H is strictly lower in 13/60, and P is strictly lower in 8/60 (Wilcoxon signed-rank on $\log_{10}(\mathrm{P}) - \log_{10}(\mathrm{H})$, $p = 0.0386$, $n = 60$, only 21 of 60 pairs non-tied). On **iterations needed to first reach the paper's own success threshold** ($R^2 \geq 0.9999$), H is never slower: 42/59 pairs tie at 1× (both arms cross the threshold on their very first observed iteration), H is strictly faster in 15/59, and P is strictly faster in 2/59 (Wilcoxon $p = 4.96 \times 10^{-4}$, $n = 59$; one pair excluded because P never reached the threshold). Restricted to the 17 pairs where the two arms actually differ, the geometric-mean speedup is 4.07× (median 4.0×, max 31×, range 0.64×–31×); pooled across all 59 pairs including ties it is 1.50×.

In short: with a real `r2_threshold=0.9999` and `niterations=1000` budget, both arms recover essentially every Nguyen-12 equation to the same tiny loss by the time each run ends, so a raw final-loss comparison mostly measures ties, not a difference. The clearer, less confounded signal is in *how many outer iterations it takes to first cross the threshold*: warm-start is never worse there and is often several times faster on the equations where it makes any difference at all. We still recommend threshold-crossing speed as the primary "does warm-start help" metric, both because it is the more informative of the two on this data and because it remains the right metric in general (final loss trivially favors whichever arm is allowed to keep iterating longer).

Data-quality note for anyone reusing these figures. The warm-start arm fell back to an unmodified cold-PySR run (`effective_method = pysr_cold_copy`) on 2 of 60 equation–seed pairs (seed 42 / N10, seed 777 / N10) because no LLM candidate was returned; the pipeline still counts these as H results, which is conservative against H rather than for it.

Pipeline fix applied. The previous version of this report flagged a display-only bug in `_ptraj_plot_aggregate_winloss`: the pooled final-loss win/loss scatter's title read "12 seed(s)" instead of "5 seed(s)" because the tuple-unpacking used the equation-ID field instead of the seed field for that one count, at `generate_figures.py` line 2089. This has been fixed (swapping which tuple position is unpacked) and all figures below, including Fig. 21, were regenerated from the corrected script. The fix only changes that one title string — the underlying win/loss/tie counts, p-values, and every other number in this report are identical to the previous version, since the bug never touched the statistics themselves, only how the seed count was displayed on one plot.

Contents

1 Methodology

Each seed file records, per Nguyen-12 equation and per arm (H = LLM warm-start hybrid, P = cold PySR), a polled trajectory of (iteration, best_loss, elapsed_seconds) observations, plus a `trajectory_summary` with `final_best_loss` and `first_threshold_iteration` (first iteration at which $R^2 \geq 0.9999$ was observed). All five runs share the same configuration: `niterations=1000`, `populations=30`, `r2_threshold=0.9999`, `trajectory_unit=outer_iteration`, deterministic, serial. For each seed we produce:

- **Overview** — the two arms' loss trajectories aggregated across all 12 equations into a single median curve per arm (H, P), rank-aligned by observation order with a 25th–75th percentile band, on a shared log-loss axis.
- **Per-equation grid** — a 3×4 grid of subplots, one per equation, each on its own y-axis, showing the real (unaggregated) H and P loss trajectories, annotated with which arm reached the lower final loss and which reached the $R^2 \geq 0.9999$ threshold first.
- **Win/loss scatter** — P's final loss (x) vs. H's final loss (y), log–log, per equation; points below the $y = x$ line are equations where H ended with a lower loss.
- **Speedup bar** — for equations where both arms reached the threshold, the ratio (P's threshold iteration) / (H's threshold iteration); bars above $1\times$ favor H.

After processing all seed files, the same two paired comparisons (final loss, and threshold-crossing speedup) are pooled across every seed and equation and tested with a paired Wilcoxon signed-rank test (`scipy.stats.wilcoxon`), producing the two cross-seed figures in Section 3.

2 Per-Seed Results

2.1 Seed 42

Final loss: H better in 1/12, P better in 2/12, tie in 9/12. Threshold speed: H faster in 3/12, P faster in 0/12, tie in 9/12. Both arms fell back to `pysr_cold_copy` for H on N10 (no LLM candidate returned); the largest single speedup in the whole 5-seed dataset is here, N12 at $31\times$ (H threshold iteration 1 vs. P's 31).

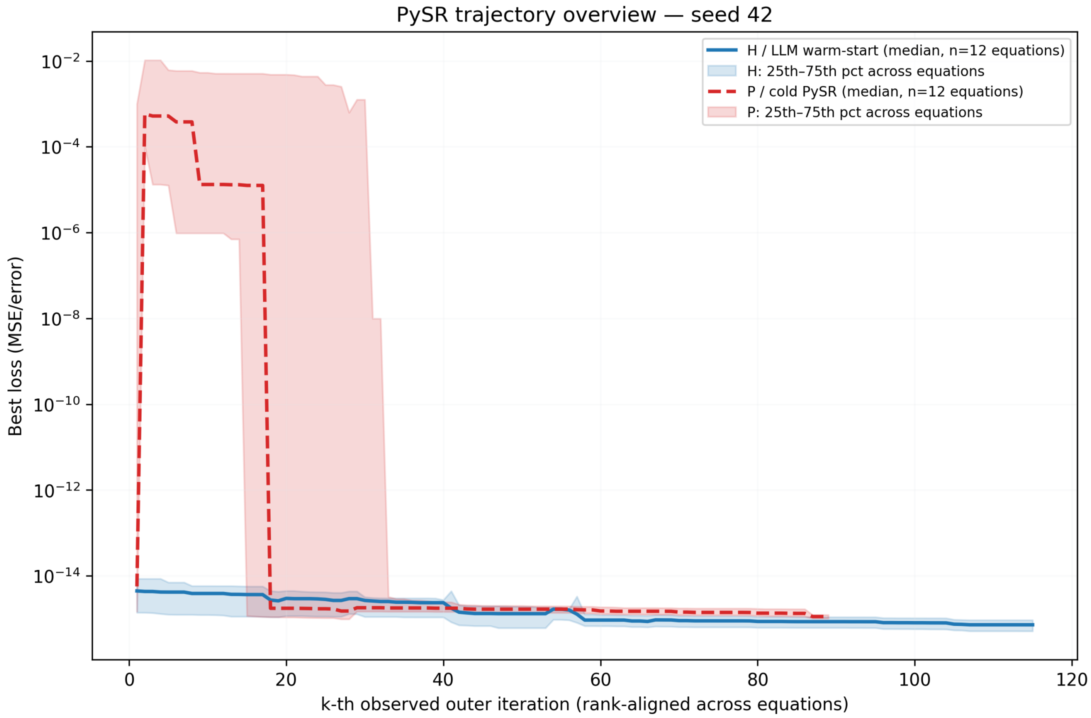

Figure 1: Seed 42 — overview (median H vs. P loss trajectory, rank-aligned, IQR band).

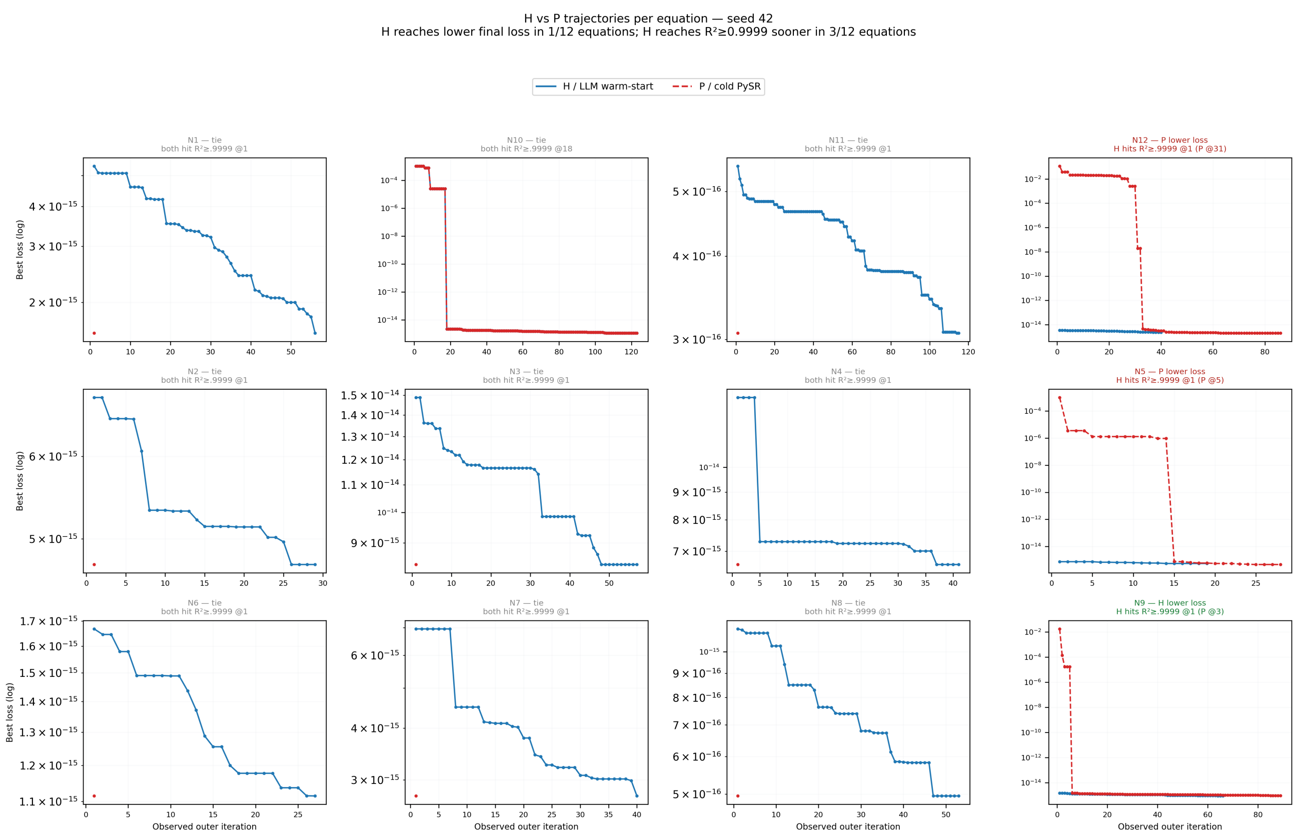

Figure 2: Seed 42 — per-equation trajectory grid (real, unaggregated H/P curves).

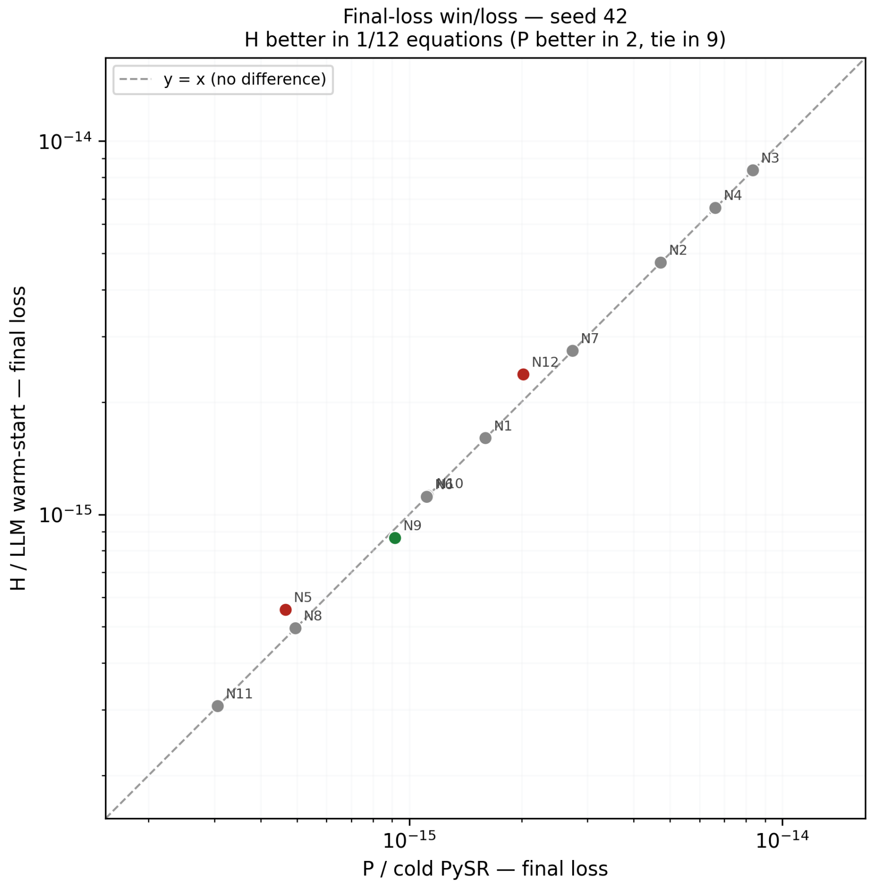

Figure 3: Seed 42 — final-loss win/loss scatter.

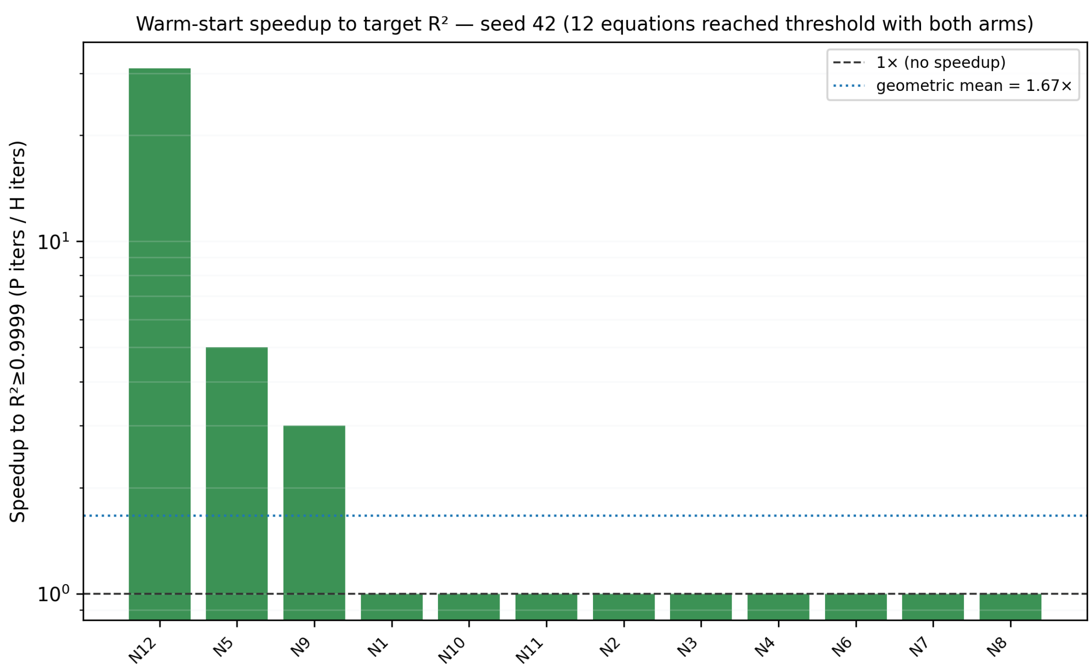

Figure 4: Seed 42 — per-equation speedup to $R^2 \geq 0.9999$.

2.2 Seed 99

Final loss: H better in 2/12, P better in 0/12, tie in 10/12. Threshold speed: H faster in 1/12, P faster in 0/12, tie in 11/12. The most lopsided seed in H's favor on final loss (P never wins outright here), though the threshold-crossing margin is modest — most equations tie at 1×.

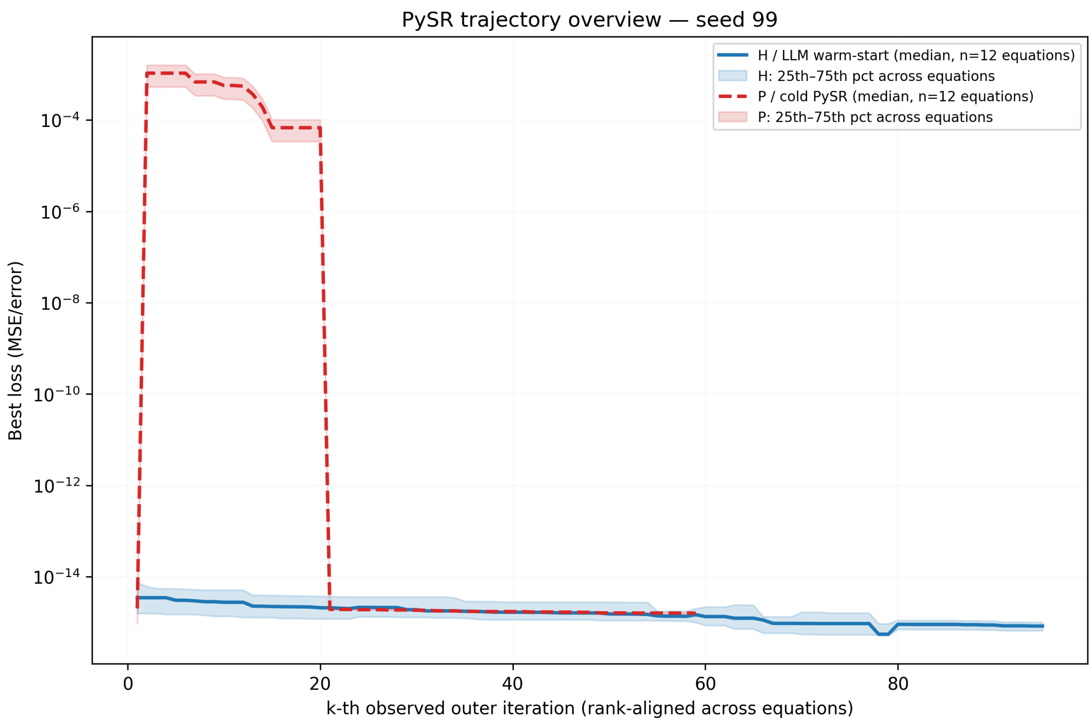

Figure 5: Seed 99 — overview (median H vs. P loss trajectory, rank-aligned, IQR band).

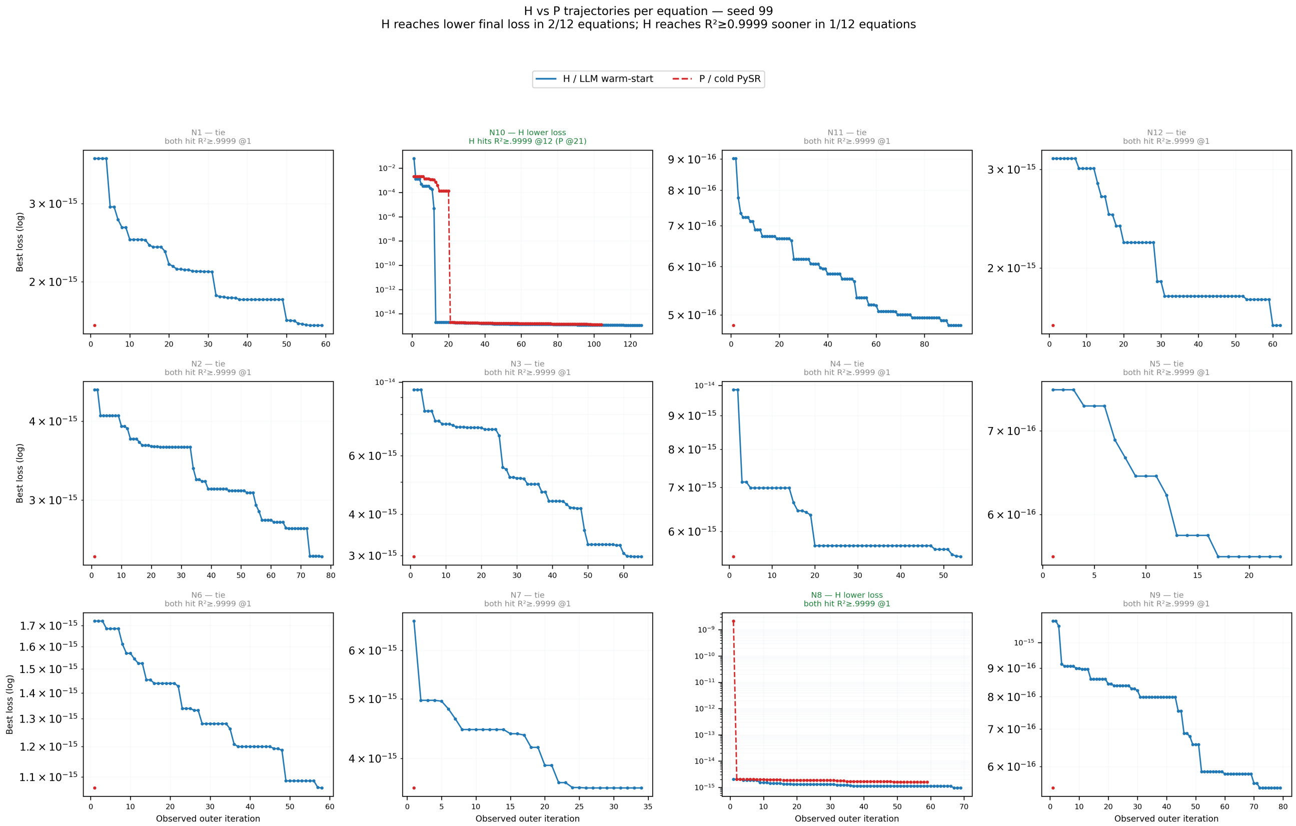

Figure 6: Seed 99 — per-equation trajectory grid (real, unaggregated H/P curves).

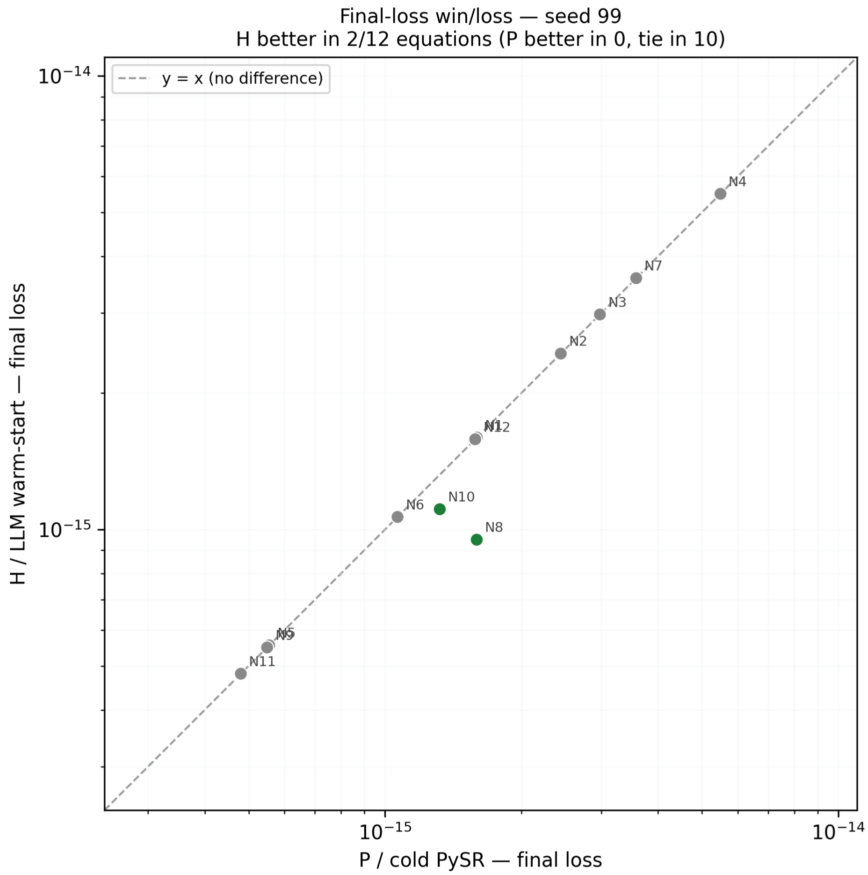

Figure 7: Seed 99 — final-loss win/loss scatter.

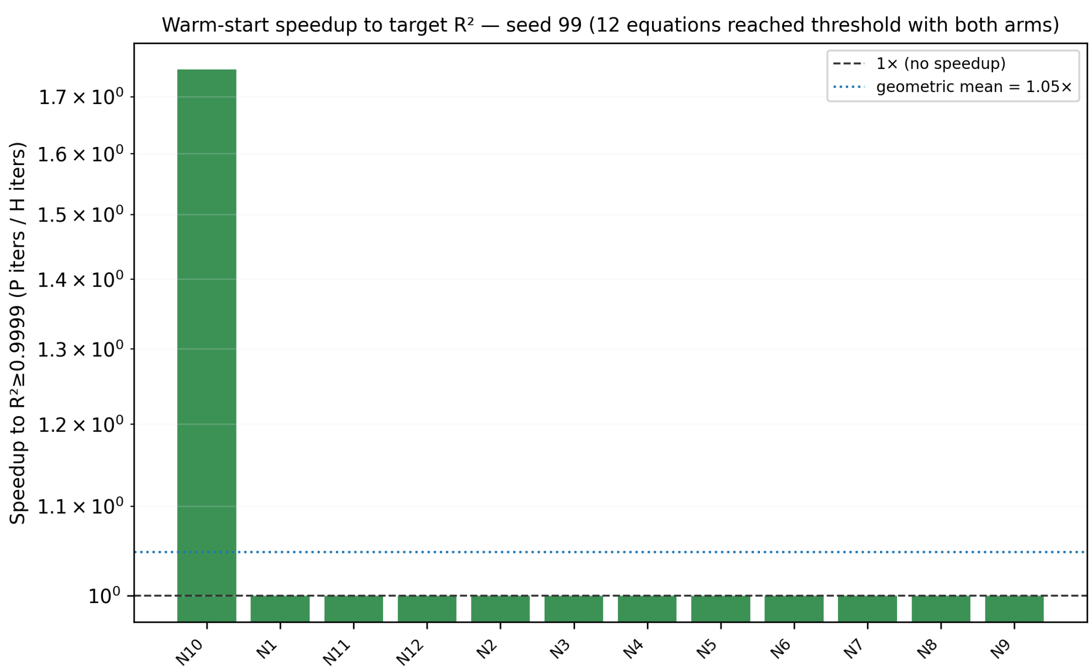

Figure 8: Seed 99 — per-equation speedup to $R^2 \geq 0.9999$.

2.3 Seed 123

Final loss: H better in 2/12, P better in 1/12, tie in 9/12. Threshold speed: H faster in 2/12, P faster in 1/12, tie in 9/12. This is the only seed where P is ever faster to threshold (N10: P reaches it at iteration 14 vs. H's 22) despite H's final loss still being competitive overall.

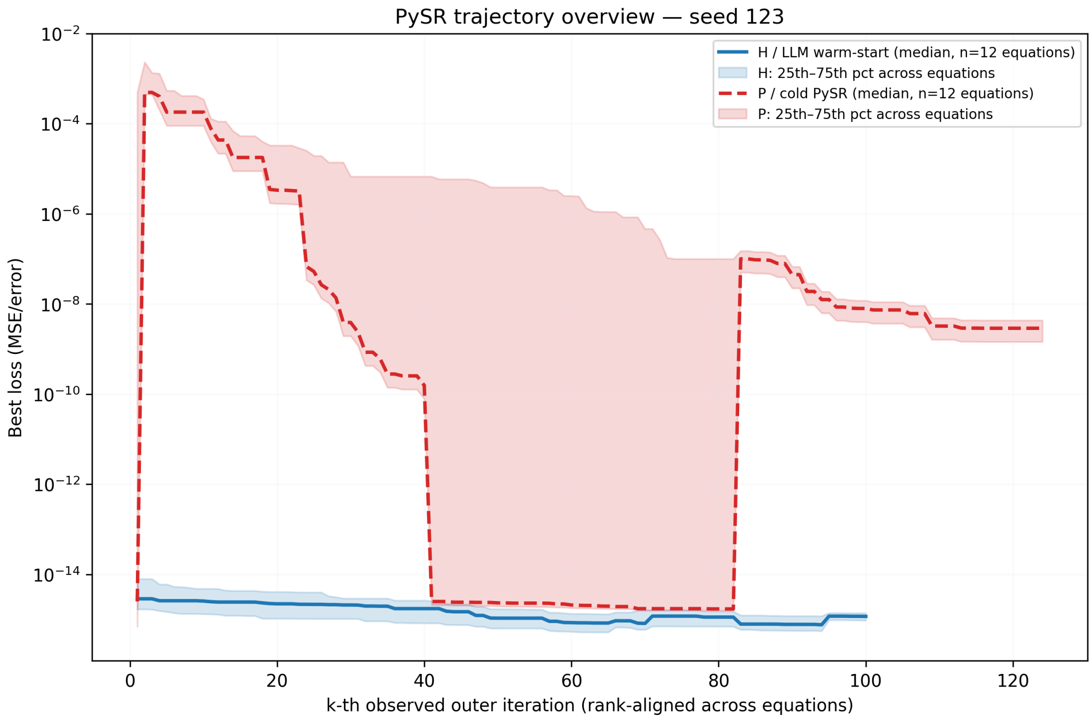

Figure 9: Seed 123 — overview (median H vs. P loss trajectory, rank-aligned, IQR band).

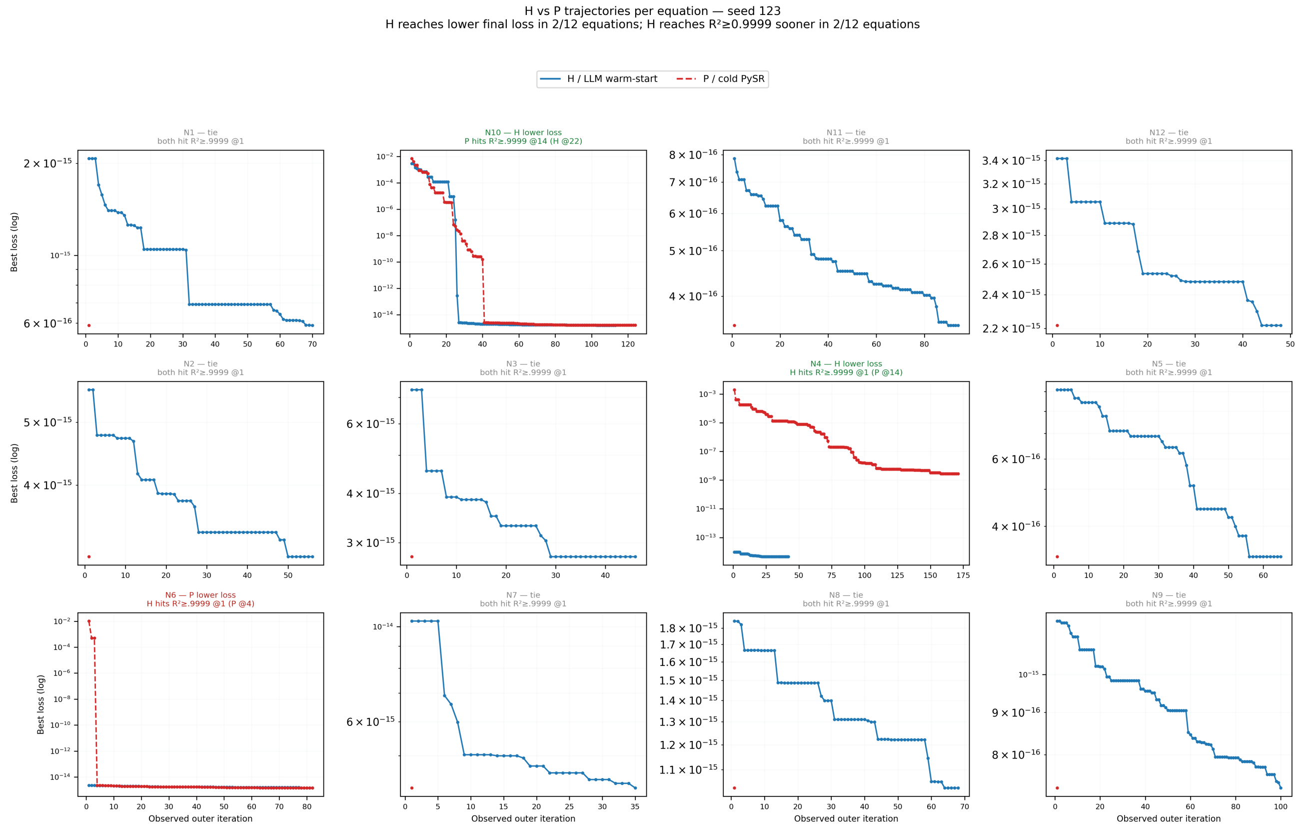

Figure 10: Seed 123 — per-equation trajectory grid (real, unaggregated H/P curves).

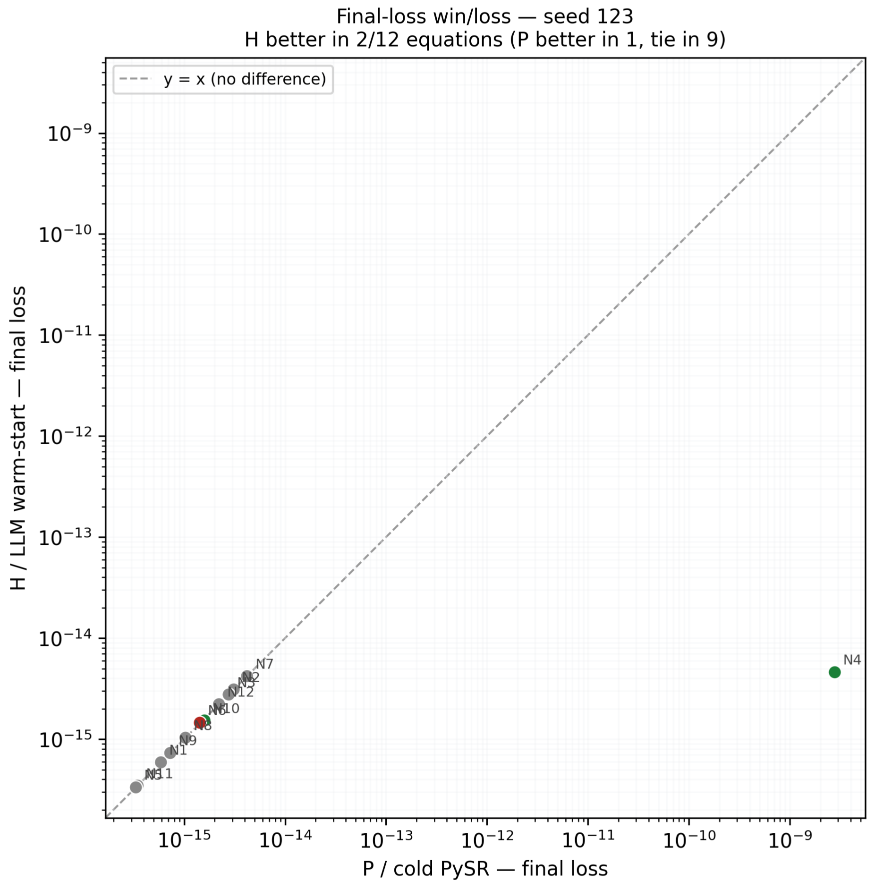

Figure 11: Seed 123 — final-loss win/loss scatter.

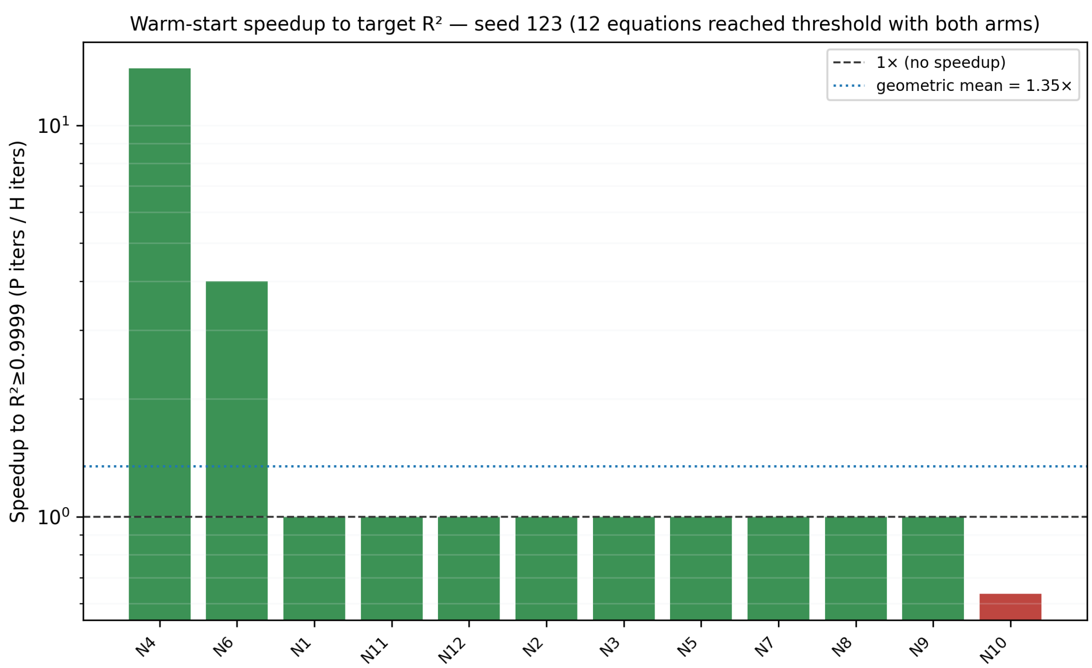

Figure 12: Seed 123 — per-equation speedup to $R^2 \geq 0.9999$.

2.4 Seed 777

Final loss: H better in 4/12, P better in 2/12, tie in 6/12 (only 11 pairs have both a valid threshold observation, since P never reaches $R^2 \geq 0.9999$ on N12). Threshold speed (11 usable pairs): H faster in 5/11, P faster in 0/11, tie in 6/11. The N10 pair again falls back to `pysr_cold_copy` for H, as in seed 42.

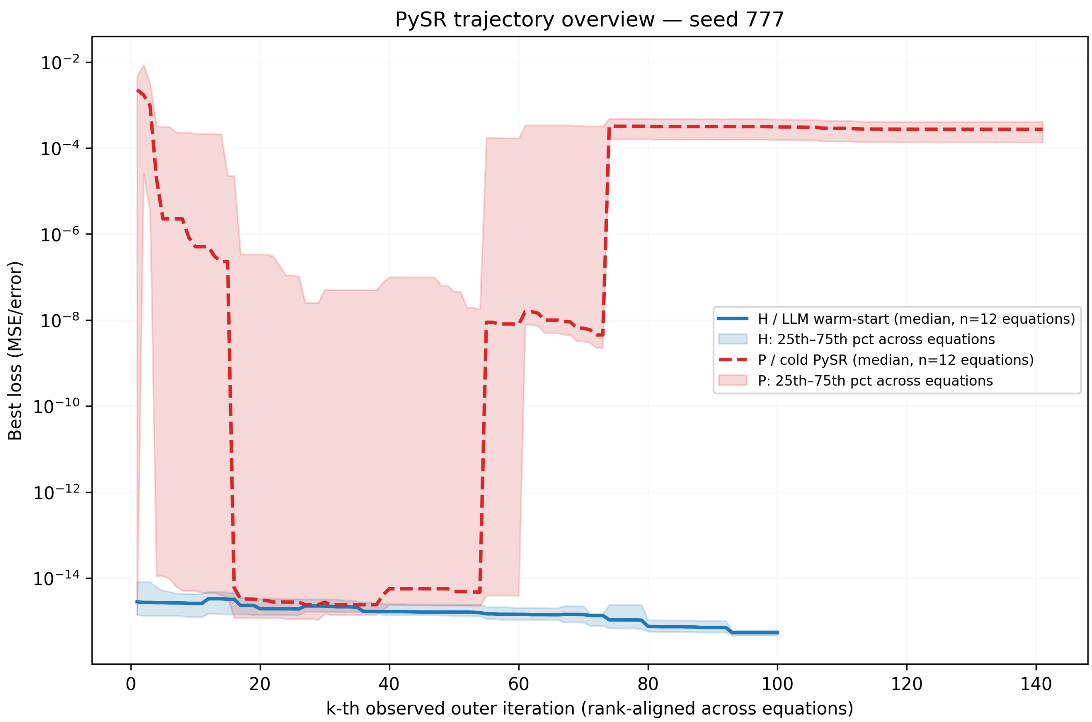

Figure 13: Seed 777 — overview (median H vs. P loss trajectory, rank-aligned, IQR band).

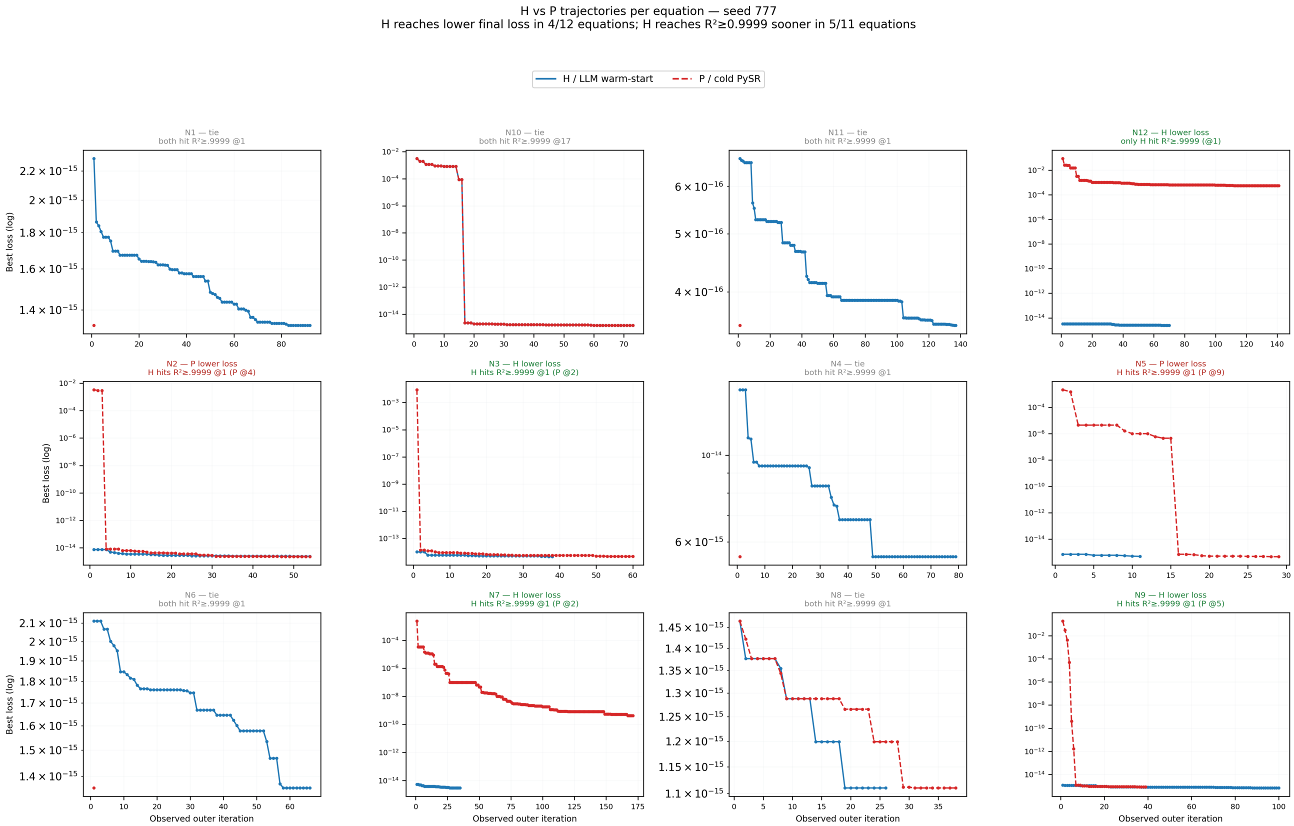

Figure 14: Seed 777 — per-equation trajectory grid (real, unaggregated H/P curves).

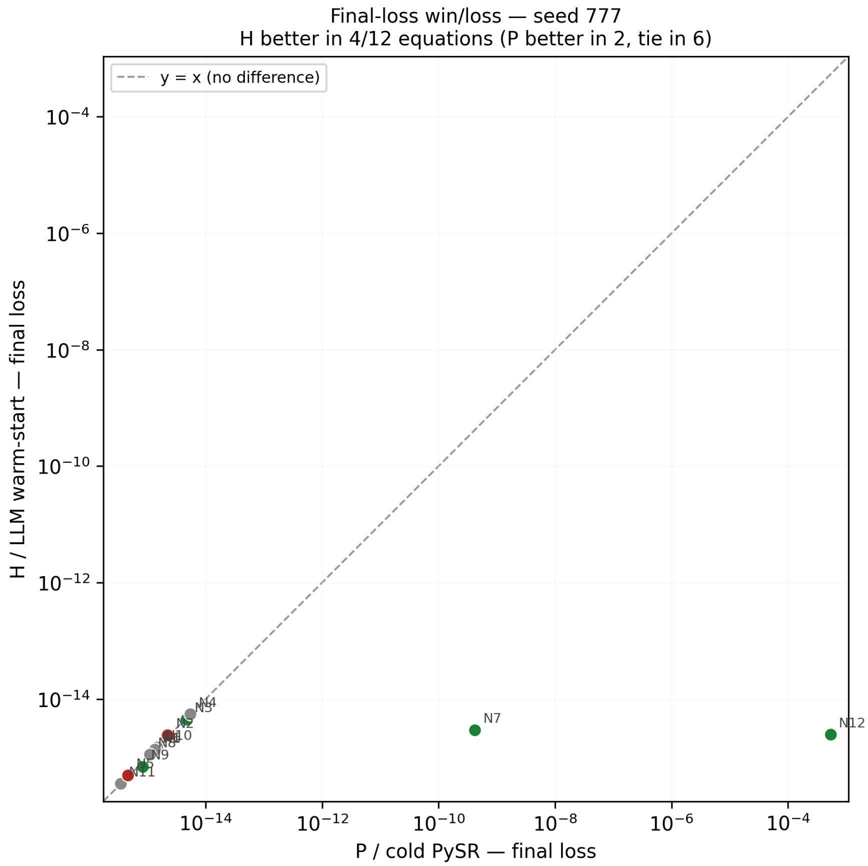

Figure 15: Seed 777 — final-loss win/loss scatter.

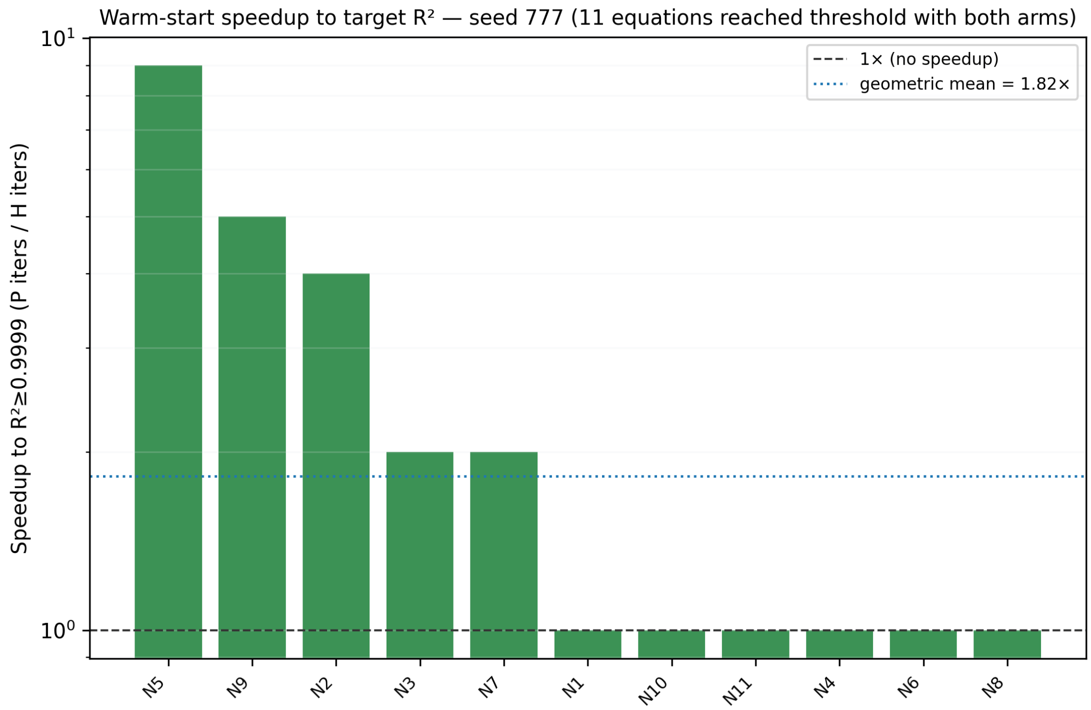

Figure 16: Seed 777 — per-equation speedup to $R^2 \geq 0.9999$.

2.5 Seed 2024

Final loss: H better in 4/12, P better in 3/12, tie in 5/12. Threshold speed: H faster in 4/12, P faster in 1/12, tie in 7/12. The most active seed for both metrics — the fewest ties of the five, with both arms winning at least a few equations outright.

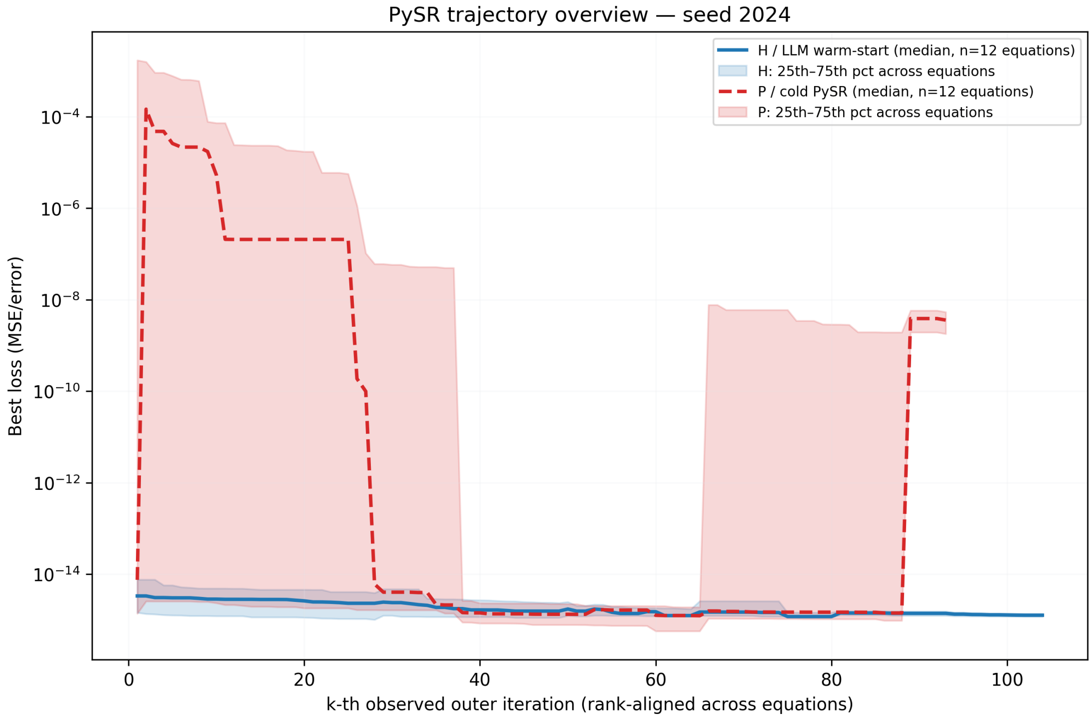

Figure 17: Seed 2024 — overview (median H vs. P loss trajectory, rank-aligned, IQR band).

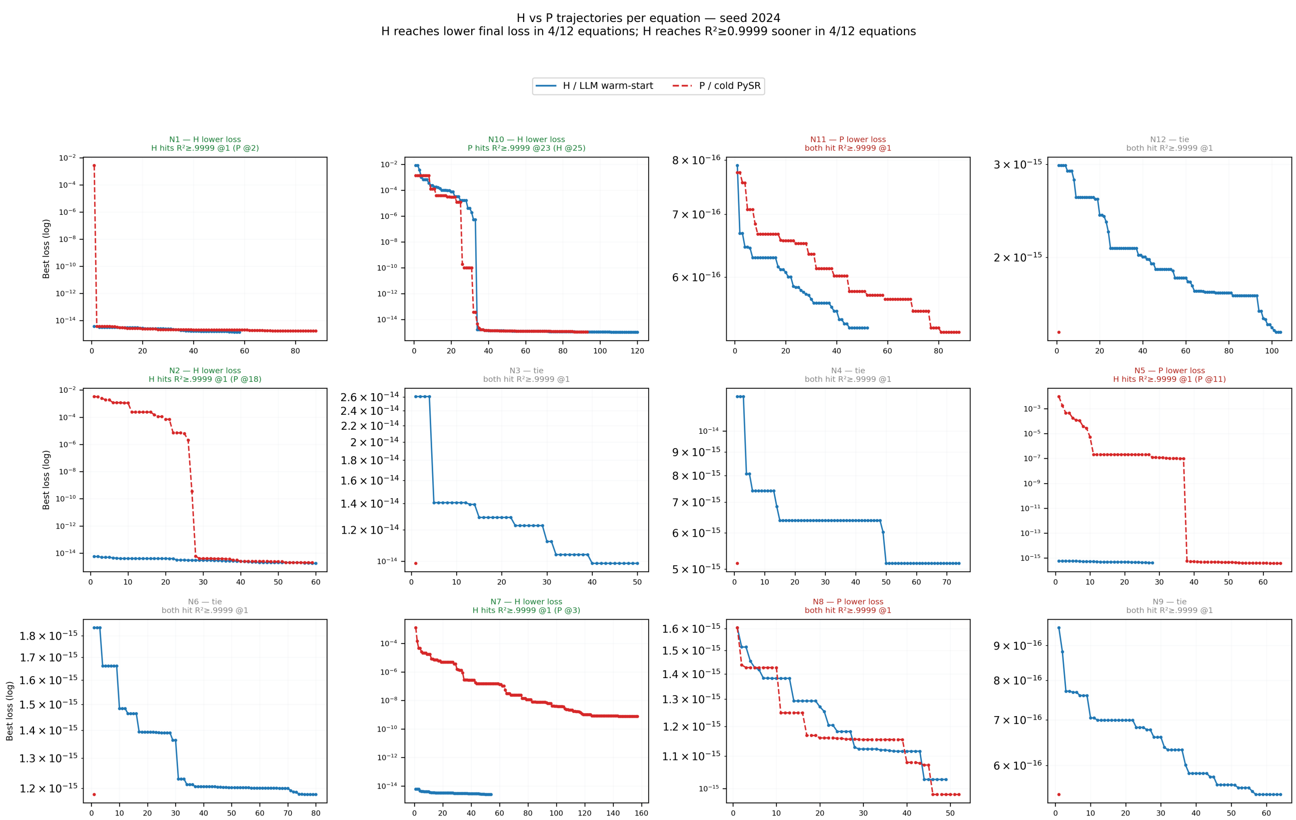

Figure 18: Seed 2024 — per-equation trajectory grid (real, unaggregated H/P curves).

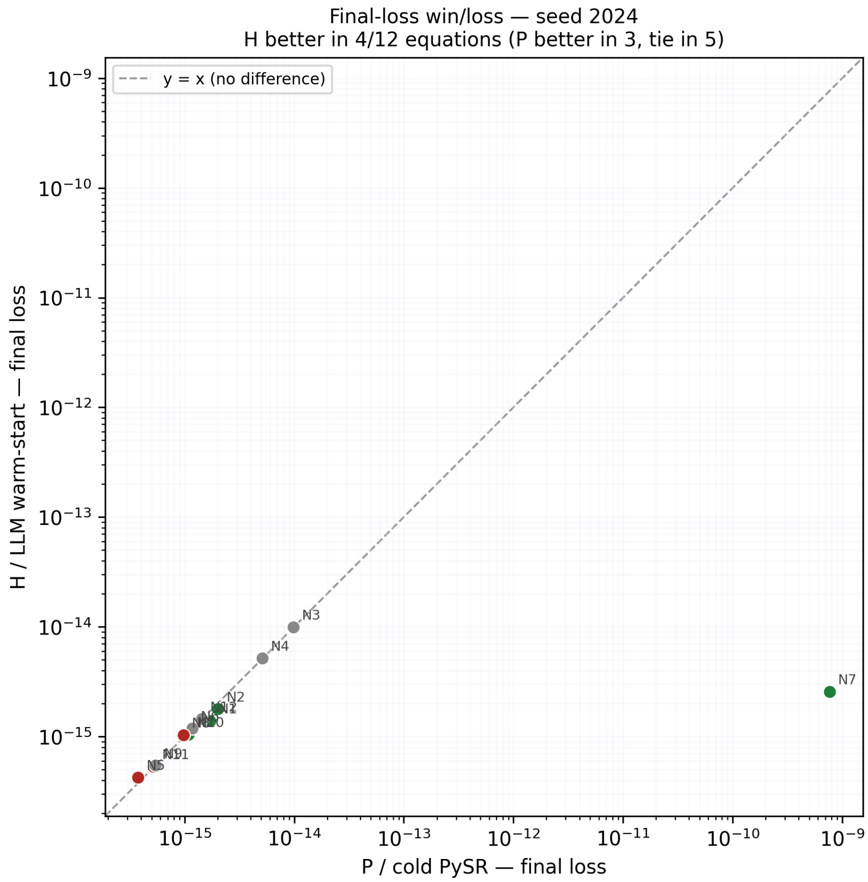

Figure 19: Seed 2024 — final-loss win/loss scatter.

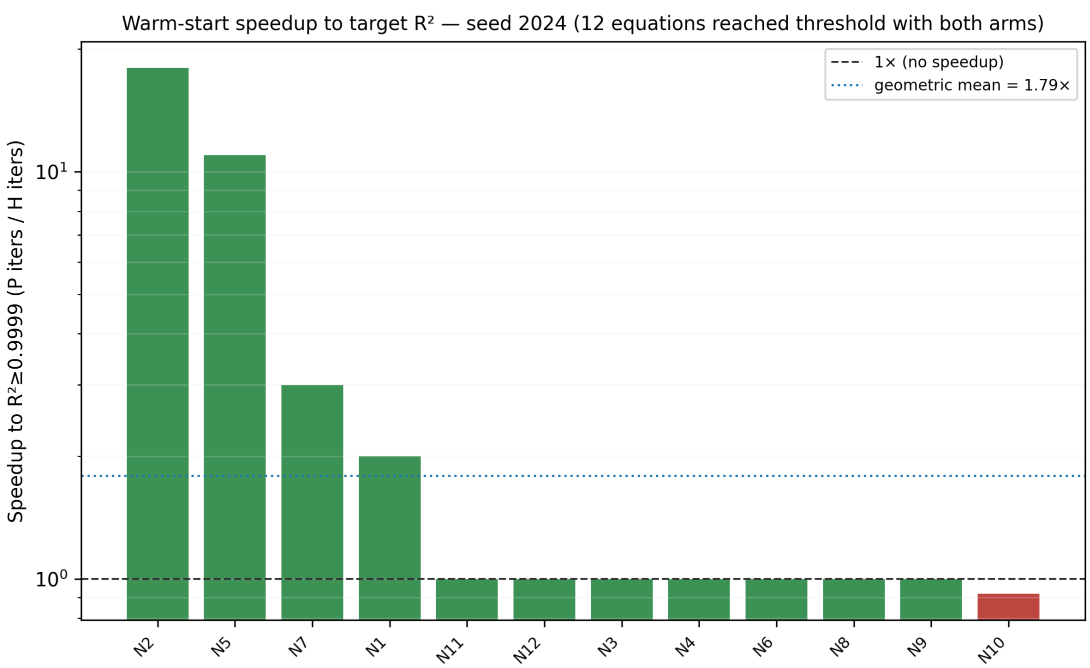

Figure 20: Seed 2024 — per-equation speedup to $R^2 \geq 0.9999$.

3 Cross-Seed Aggregate

All 5 seeds × 12 equations pooled (60 equation–seed pairs for final loss; 59 for threshold-crossing speed, since P never reaches threshold on seed 777 / N12).

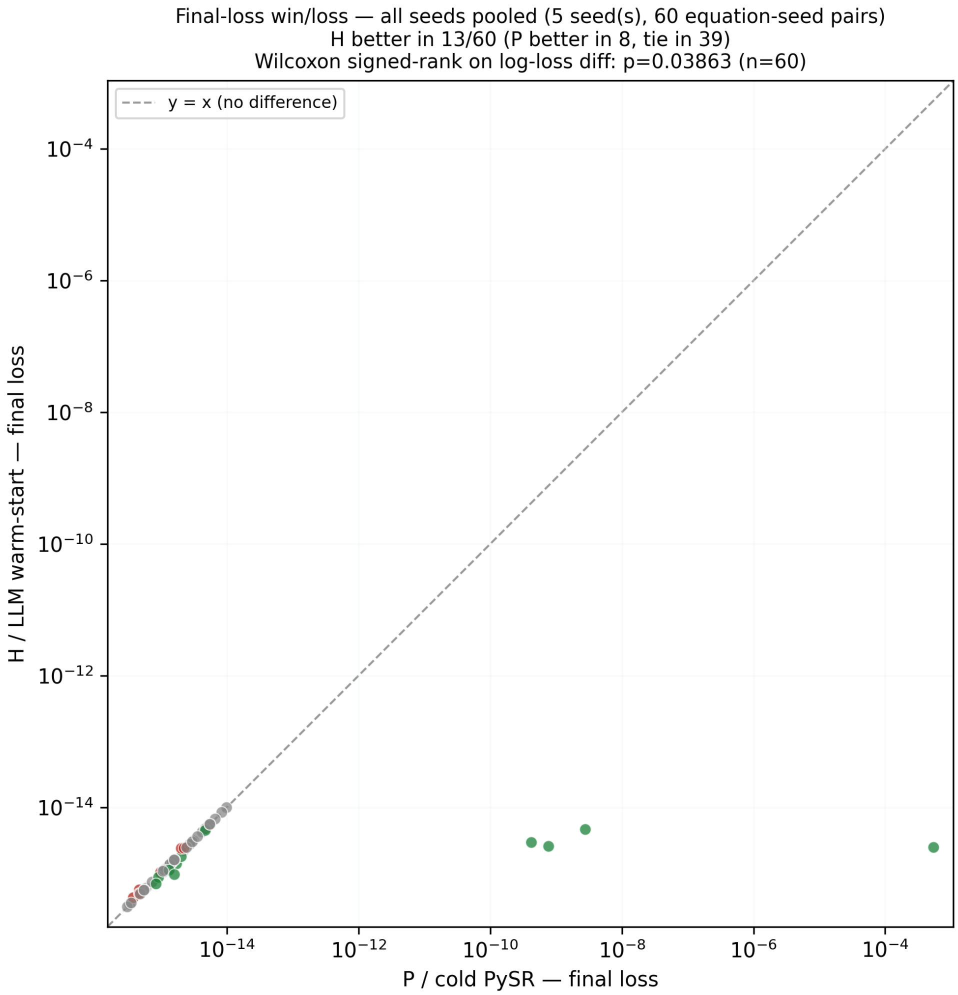

Figure 21: Pooled final-loss win/loss across all 5 seeds, with paired Wilcoxon signed-rank test on $\log_{10}(P) - \log_{10}(H)$. Title now correctly reads "5 seed(s)" following the `generate_figures.py` fix described in the abstract.

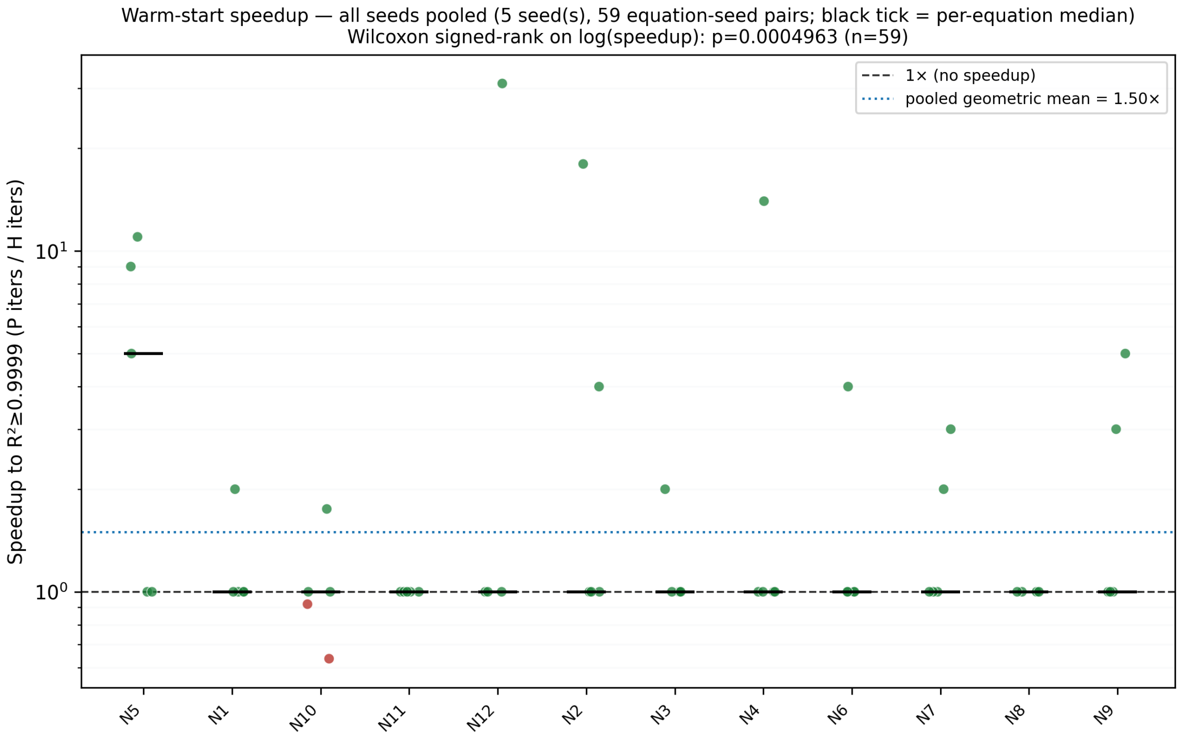

Figure 22: Pooled speedup to $R^2 \geq 0.9999$ across all 5 seeds (jittered points per equation, black tick = per-equation median), with paired Wilcoxon signed-rank test on log(speedup).

Table 1: Cross-seed paired Wilcoxon signed-rank tests, as printed by the pipeline (`exp3_traj_ALLSEEDS_stats_summary.txt`).

Metric	n pairs	n non-tied	Median diff.	Statistic	p-value	Favors H / P
$\log_{10}$(P) − $\log_{10}$(H) final loss	60	21	0	56	0.0386	13 / 8
log(P iters/H iters) to threshold	59	17	0	3	4.96×10^{-4}	15 / 2

The median diff. is 0 for both metrics because a majority of pairs are exact ties (both arms end at the same final loss, or both cross the threshold on iteration 1); `scipy.stats. wilcoxon` drops zero-differences before ranking, so the reported statistic and p-value are computed on the 21 and 17 non-tied pairs respectively, while n in the table is the full pair count for context. Restricted to the 17 non-tied threshold-speed pairs, the geometric-mean speedup is 4.07× (median 4.0×, max 31×, all in H's favor except two: seed 123/N10 at 0.64× and seed 2024/N10 at 0.92×); pooled across all 59 pairs including ties the geometric mean is 1.50×.

4 Conclusions

1. **With a real, generous iteration budget, the two arms mostly tie on final loss — the interesting signal is in speed, not endpoint quality.** 39 of 60 equation–seed pairs end in an exact final-loss tie, both arms having driven the loss to $\sim 10^{-15}$–10^{-16}. That is a good outcome for the benchmark (both methods solve almost everything) but it means a bare final-loss comparison is close to uninformative here: the real question — and the one warm-start is meant to answer — is how many iterations it takes to get there.

2. **On that question, warm-start is never worse in the aggregate and is often substantially faster.** 15 of 59 threshold-crossing pairs favor H outright (up to 31×), only 2 favor P (at 0.64× and 0.92×, i.e. mild slowdowns, both on the same equation, N10), and 42 tie at 1× because both arms happened to cross the threshold on their first observed iteration. The paired Wilcoxon test on the non-tied pairs is significant at $p = 4.96 \times 10^{-4}$.

3. **Two equation–seed pairs (seed 42/N10, seed 777/N10) had H silently fall back to a cold-PySR run** because the LLM warm-start proposal was rejected or not returned (`effective_method = pysr_cold_copy`). Both of those pairs are counted as H results in every statistic above, which is conservative (it can only hurt H's numbers, never help them) — but it also means the true warm-start effect, restricted to runs where a candidate was actually used, is at least as strong as what is reported here, and worth recomputing separately if this number goes in the paper.

4. **The pooled final-loss scatter's seed-count display bug has been fixed.** The previous report round flagged a tuple-unpacking mistake in `_ptraj_plot_aggregate_winloss` that read the equation-ID field instead of the seed field when computing the "N seed(s)" count in that one plot's title (it read "12" instead of "5"). The fix is a one-line change (swap which tuple position is unpacked, matching the three other places in the file that already unpack this tuple correctly), applied directly to `generate_figures.py` line 2089, and all 82 figures in this report — including Fig. 21 — were regenerated from the corrected script. As expected for a display-only fix, every statistic in this report (win/loss/tie counts, p-values, speedups) is unchanged from the previous version. This was the same class of issue as the separately-flagged Nguyen-12 success-count arithmetic problem in `01_nguyen12_success_count_arithmetic.md` (three disagreeing success-count denominators for one table): a value that should be computed once from a single source field was instead recomputed from the wrong field in one place. We recommend applying the same fix there — compute each summary number from an unambiguous, explicitly-named source field, and have the script itself export/print the number used in each caption or table cell, rather than trusting a second ad hoc computation to agree with the first.